# Data Communication & Computer Networks (CN)

## Topic 3 of 7 : Addressing (MAC, IPv4, IPv6, Port, DNS)

**81 MCQs** — Repo Exam Questions + Forouzan — bilingual (English + Hindi), exam-ready

Reference Book: **Data Communications and Networking — Behrouz A. Forouzan**

Prepared for: **RPSC Programmer • ACP • RSMSSB IA • SA • DD • BCI • SCI • REET • RAS • DSSSB • KVS • NVS • HTET • SSC CGL/CHSL • RRB • IBPS • UPSC & all Rajasthan + Central Govt. Exams (Computer Awareness)**

Total Questions: **81** MCQs + **27** Nukte rapid-fire revision points

Question Type: MCQ — 4 options (A, B, C, D) per question, in English + Hindi

Layout: Left = **question** | Right = **explanation** (green): ✔ **why the correct answer is right** + ★ **Nukta** (key exam point)

Language rule: Hindi is the second language; **technical terms stay in English** (topology, protocol, jitter…)

Verification: Every answer cross-verified **4×** (Forouzan + official answer keys of Subject Test / Unit Test Solutions). Wrong options verified internally.

## Syllabus Coverage — sub-topic wise order


**1.** Section A — Address Levels & MAC Address (Physical Address) — Q1 - Q10

**2.** Section B — IPv4 Addressing (Format, Classes & Special Addresses) — Q11 - Q39

**3.** Section C — Subnetting & CIDR — Q40 - Q61

**4.** Section D — IPv6 Addressing — Q62 - Q69

**5.** Section E — Port Numbers, DNS/URL & Address Resolution (ARP) — Q70 - Q81


**How to read:** Left side: question + 4 options (try it yourself first). Right side (green): ✔ correct answer with why it is right, and ★ **Nukta** — the one-line exam tip to remember. Practice in 2 passes: first read question + options only, then check the green column.

**System:** generated by **tools/cn_build_topic.py** from **tools/data/topic3_data.py**. To add more questions later: drop PDFs into **Computer Network/Incoming/** and run **tools/cn_pipeline.py** — new questions auto-classify topic-wise. See **Computer Network/README.md**.

## Section A — Address Levels & MAC Address (Physical Address)

Forouzan's four address levels; the 48-bit burned-in MAC address, unicast/multicast/broadcast MAC.

**Q1.** Which addresses are used in an internet employing the TCP/IP protocols?

TCP/IP protocols वाले internet में कौन-से addresses उपयोग होते हैं?

**(A)** Physical address and logical address **(B)** Port address
**(C)** Specific address **(D)** All of these

[Source: LMT MCQ bank (Computer Communication Networks)]

✔ **Correct: (D) All of these** — TCP/IP communication uses all four levels of addresses: physical (MAC), logical (IP), port, and specific (socket = MAC+IP+port) — depending on the layer.

TCP/IP communication में चारों levels के addresses लगते हैं: physical (MAC), logical (IP), port, और specific (socket = MAC+IP+port) — layer के अनुसार।

★ **Nukta:** Forouzan's 4 address levels: Physical(48-bit MAC) at data link, IP(32-bit) at network, Port(16-bit) at transport, Specific (socket) at application.

Forouzan की 4 address levels: Physical(48-bit MAC) data link पर, IP(32-bit) network पर, Port(16-bit) transport पर, Specific (socket) application पर।

**Q2.** What is the size (length) of a MAC address?

MAC address का size (लंबाई) कितना होता है?

**(A)** 42 bits **(B)** 48 bits
**(C)** 24 bits **(D)** 36 bits

[Source: RPSC Programmer 2024 Paper-1 (official key) · also BCI_SCI ST-7 / Sanfoundry / InstrumentationTools]

✔ **Correct: (B) 48 bits** — A MAC (Media Access Control) address is 48 bits (6 bytes) long, usually written as 12 hex digits like 00:1A:2B:3C:4D:5E.

MAC (Media Access Control) address 48 bits (6 bytes) का होता है, जिसे आमतौर पर 12 hex digits में लिखा जाता है — जैसे 00:1A:2B:3C:4D:5E।

★ **Nukta:** MAC = 48 bits (6 bytes / 12 hex digits). IPv4 = 32 bits, IPv6 = 128 bits, Port = 16 bits. Ye char sizes exam में बार-बार आते हैं।

MAC = 48 bits (6 bytes / 12 hex digits)। IPv4 = 32 bits, IPv6 = 128 bits, Port = 16 bits। ये चार sizes exam में बार-बार आते हैं।

**Q3.** A MAC address is also called a _____ address.

MAC address को _____ address भी कहा जाता है।

**(A)** Physical address **(B)** Logical address
**(C)** Source address **(D)** Destination address

[Source: Repo: CN Topic Test-01 (UNIT TEST-1) · also BCI_SCI ST-7]

✔ **Correct: (A) Physical address** — The MAC address is permanently burned into the NIC — it is the physical (hardware) address of the device, unlike the logical IP address which is configured by software.

MAC address NIC में permanently burn होता है — यह device का physical (hardware) address है, logical IP address के विपरीत जो software से configure होता है।

★ **Nukta:** MAC = physical/hardware address (burned-in, 48-bit). IP = logical address (configured). Both change hands at different layers.

MAC = physical/hardware address (burned-in, 48-bit)। IP = logical address (configured)।

**Q4.** The address used in the Data Link Layer is called:

Data Link Layer में उपयोग होने वाले address को क्या कहते हैं?

**(A)** Logical address **(B)** Port address
**(C)** MAC address **(D)** IP address

[Source: Repo: CN Topic Test-04 (UNIT TEST-4)]

✔ **Correct: (C) MAC address** — The data link layer uses the MAC (physical/hardware) address for frame delivery on the local link.

Data link layer local link पर frame delivery के लिए MAC (physical/hardware) address उपयोग करती है।

★ **Nukta:** Layer-address map again: Data link→MAC, Network→IP, Transport→Port, Application→socket.

Layer-address map: Data link→MAC, Network→IP, Transport→Port, Application→socket।

**Q5.** Which of the following is a static hardware address?

निम्न में से कौन-सा static hardware address है?

**(A)** IP address **(B)** MAC address
**(C)** Port number **(D)** Domain name

[Source: Repo: BCI_SCI Subject Test-7 (BCI/SCI Batch)]

✔ **Correct: (B) MAC address** — The MAC address is static — it is assigned by the manufacturer and does not normally change; IP addresses, ports and domain names are all dynamic/logical.

MAC address static होता है — यह manufacturer द्वारा दिया जाता है और सामान्यतः नहीं बदलता; IP addresses, ports और domain names सब dynamic/logical हैं।

★ **Nukta:** Static = burned into hardware (MAC). Dynamic/logical = configured (IP via DHCP, port per connection, name via DNS).

Static = hardware में burned (MAC)। Dynamic/logical = configure होते हैं (IP via DHCP, port per connection, name via DNS)।

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**Q6.** Which address identifies a unique device on a local network segment?

Local network segment पर कौन-सा address एक unique device की पहचान करता है?

**(A)** IP address **(B)** MAC address
**(C)** Gateway **(D)** Subnet mask

[Source: Repo: BCI_SCI Subject Test-7 (BCI/SCI Batch)]

✔ **Correct: (B) MAC address** — Within a local network segment (LAN), frames are delivered using the MAC address — the unique 48-bit hardware address of each NIC.

Local network segment (LAN) के भीतर frames की delivery MAC address से होती है — हर NIC का unique 48-bit hardware address।

★ **Nukta:** L2 delivery (same network) = MAC. L3 delivery (across networks) = IP. Exam loves this pair.

L2 delivery (same network) = MAC। L3 delivery (across networks) = IP। Exam इस जोड़ी को बहुत पूछता है।

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**Q7.** What type of MAC address is FF:FF:FF:FF:FF:FF?

FF:FF:FF:FF:FF:FF किस प्रकार का MAC address है?

**(A)** Unicast **(B)** Multicast
**(C)** Broadcast **(D)** Loopback

[Source: Repo: CN Topic Test 5 (UNIT TEST-5)]

✔ **Correct: (C) Broadcast** — FF:FF:FF:FF:FF:FF is the broadcast MAC address — a frame sent to it is received by every device on the LAN segment.

FF:FF:FF:FF:FF:FF broadcast MAC address है — इसे भेजा frame LAN segment के हर device को मिलता है।

★ **Nukta:** MAC broadcast = FF:FF:FF:FF:FF:FF (all 48 bits 1). Unicast = first byte's last bit 0; multicast = first byte's last bit 1.

MAC broadcast = FF:FF:FF:FF:FF:FF (सभी 48 bits 1)। Unicast = first byte का अंतिम bit 0; multicast = first byte का अंतिम bit 1।

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**Q8.** An Ethernet destination address 0A-01-12-03-04-05 is a:

Ethernet destination address 0A-01-12-03-04-05 कौन-सा address है?

**(A)** Unicast address **(B)** Multicast address
**(C)** Broadcast address **(D)** All of these

[Source: Repo: CN Topic Test-04 (UNIT TEST-4)]

✔ **Correct: (A) Unicast address** — The first byte 0A = 0000 1010 has its last bit (I/G bit) = 0, which marks a unicast address; multicast addresses have this bit = 1.

पहला byte 0A = 0000 1010 का अंतिम bit (I/G bit) = 0 है, जो unicast address दर्शाता है; multicast में यह bit = 1 होता है।

★ **Nukta:** I/G bit (last bit of first octet): 0 = unicast, 1 = multicast. All-ones (FF:FF...) = broadcast.

I/G bit (पहले octet का अंतिम bit): 0 = unicast, 1 = multicast। All-ones (FF:FF...) = broadcast।

**Q9.** Consider the statements: S1: Every computer or Internet device has a unique MAC address. S2: Every computer or Internet device has a unique IP address. Which is correct?

कथन देखिए: S1: हर computer या Internet device का unique MAC address होता है। S2: हर computer या Internet device का unique IP address होता है। कौन-सा सही है?

**(A)** S1 is true and S2 is false **(B)** Both S1 and S2 are false
**(C)** S1 is false and S2 is true **(D)** Both S1 and S2 are true

[Source: Repo: CN Topic Test-01 (UNIT TEST-1)]

✔ **Correct: (A) S1 is true and S2 is false** — S1 is true — MAC addresses are globally unique (burned in by the manufacturer). S2 is false — IP addresses are logical and can be duplicated in different private networks (and change via DHCP).

S1 सही है — MAC addresses globally unique होते हैं (manufacturer द्वारा burn किए गए)। S2 गलत है — IP addresses logical होते हैं और अलग-अलग private networks में duplicate हो सकते हैं (और DHCP से बदलते हैं)।

★ **Nukta:** MAC = permanent + globally unique. IP = logical + can repeat across private networks (NAT handles the overlap).

MAC = स्थायी + globally unique। IP = logical + private networks में दोहराया जा सकता है (NAT overlap संभालता है)।

**Q10.** Which header field is rewritten (not removed) when an IP packet passes through a router?

जब एक IP packet router से गुजरता है तो कौन-सा header field बदला (rewrite) जाता है?

**(A)** Source IP address **(B)** Destination IP address
**(C)** MAC (Media Access Control) address **(D)** Subnet mask

[Source: Repo: CN Topic Test-03 (UNIT TEST-3)]

✔ **Correct: (C) MAC (Media Access Control) address** — At each hop the router strips the old frame and writes NEW source and destination MAC addresses in the new frame; the IP addresses stay unchanged end-to-end.

हर hop पर router पुराना frame हटाकर नए frame में नए source और destination MAC addresses लिखता है; IP addresses end-to-end अपरिवर्तित रहते हैं।

★ **Nukta:** Per hop: MAC header changes. End-to-end: IP addresses stay the same. TTL also decrements (a router function).

Per hop: MAC header बदलता है। End-to-end: IP addresses same रहते हैं। TTL भी घटता है (router का काम)।

## Section B — IPv4 Addressing (Format, Classes & Special Addresses)

32-bit dotted-decimal, octets, classes A-E, default masks, loopback/APIPA/private/special addresses, NAT.

**Q11.** What is the size of each octet in an IPv4 address?

IPv4 address के प्रत्येक octet का size कितना होता है?

**(A)** 4 bits **(B)** 8 bits
**(C)** 16 bits **(D)** 32 bits

[Source: IP Addressing quiz (Studocu bank, with keys)]

✔ **Correct: (B) 8 bits** — An IPv4 address has 4 octets and each octet is 8 bits — 4 × 8 = 32 bits total.

IPv4 address में 4 octets होते हैं और हर octet 8 bits का होता है — 4 × 8 = कुल 32 bits।

★ **Nukta:** Octet = 8 bits (0-255 in decimal). 4 octets = 32 bits = dotted-decimal notation.

Octet = 8 bits (decimal में 0-255)। 4 octets = 32 bits = dotted-decimal notation।

**Q12.** An IPv4 address is of size:

IPv4 address कितने bits का होता है?

**(A)** 8 bit **(B)** 16 bit
**(C)** 32 bit **(D)** 64 bit

[Source: Networking MCQ bank (MQuestions blog)]

✔ **Correct: (C) 32 bit** — IPv4 uses 32-bit addresses written in dotted-decimal as four 0-255 octets (e.g., 192.168.1.1).

IPv4 में 32-bit addresses होते हैं जिन्हें dotted-decimal में चार 0-255 octets के रूप में लिखा जाता है (जैसे 192.168.1.1)।

★ **Nukta:** Size shortcut: MAC 48, IPv4 32, IPv6 128, Port 16 (bits).

Size shortcut: MAC 48, IPv4 32, IPv6 128, Port 16 (bits)।

**Q13.** What is the maximum number of unique IPv4 addresses?

Unique IPv4 addresses की अधिकतम संख्या कितनी है?

**(A)** 2^16 **(B)** 2^24
**(C)** 2^32 **(D)** 2^64

[Source: IP Addressing quiz (Studocu bank, with keys)]

✔ **Correct: (C) 2^32** — With 32 bits there are 2^32 (about 4.29 billion) possible IPv4 addresses — which is why IPv6 (2^128) was introduced.

32 bits के साथ 2^32 (लगभग 4.29 billion) संभावित IPv4 addresses हैं — इसीलिए IPv6 (2^128) लाया गया।

★ **Nukta:** 2^32 ≈ 4.29 billion IPv4 addresses. IPv6 = 2^128 ≈ 3.4 × 10^38.

2^32 ≈ 4.29 billion IPv4 addresses। IPv6 = 2^128 ≈ 3.4 × 10^38।

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**Q14.** An IPv4 address is divided into how many parts?

IPv4 address कितने भागों में बाँटा जाता है?

**(A)** 2 **(B)** 3
**(C)** 4 **(D)** 5

[Source: Repo: BCI_SCI Subject Test-7 (BCI/SCI Batch)]

✔ **Correct: (A) 2** — An IPv4 address has two parts: the network (prefix) part and the host (interface) part — the boundary is set by the subnet mask.

IPv4 address के दो भाग होते हैं: network (prefix) भाग और host (interface) भाग — इनकी सीमा subnet mask तय करता है।

★ **Nukta:** 2 parts = network + host. IPv4 CLASSIC division; subnetting borrows bits from host to make sub-networks.

2 parts = network + host। Subnetting में host bits से bits उधार लेकर sub-networks बनते हैं।

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**Q15.** The first octet of a Class A IP address ranges from:

Class A IP address का पहला octet किस range में होता है?

**(A)** 0-127 **(B)** 128-191
**(C)** 192-223 **(D)** 224-239

[Source: Repo: BCI_SCI Subject Test-7 (BCI/SCI Batch)]

✔ **Correct: (A) 0-127** — Class A covers first-octet values 0-127 (binary 0xxxxxxx) — though 0 and 127 are reserved, so usable Class A networks are 1-126.

Class A में पहले octet के मान 0-127 (binary 0xxxxxxx) होते हैं — हालाँकि 0 और 127 reserved हैं, इसलिए उपयोगी Class A networks 1-126 हैं।

★ **Nukta:** Class ranges by first octet: A 0-127, B 128-191, C 192-223, D 224-239 (multicast), E 240-255 (experimental).

पहले octet से Class ranges: A 0-127, B 128-191, C 192-223, D 224-239 (multicast), E 240-255 (experimental)।

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**Q16.** The first octet of a Class B IP address falls between:

Class B IP address का पहला octet किस range में आता है?

**(A)** 0-127 **(B)** 128-191
**(C)** 192-223 **(D)** 224-239

[Source: Repo: CN Topic Test-03 (UNIT TEST-3)]

✔ **Correct: (B) 128-191** — Class B uses first-octet values 128-191 (binary 10xxxxxx) with a default mask of 255.255.0.0.

Class B में पहले octet के मान 128-191 (binary 10xxxxxx) होते हैं और default mask 255.255.0.0 होता है।

★ **Nukta:** B = 128-191 (default /16). Trick: A starts 0, B starts 128, C starts 192, D starts 224, E starts 240.

B = 128-191 (default /16)। Trick: A 0 से, B 128 से, C 192 से, D 224 से, E 240 से।

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**Q17.** How many bits are used for the network portion in a Class C IP address?

Class C IP address में network भाग के लिए कितने bits उपयोग होते हैं?

**(A)** 8 **(B)** 16
**(C)** 24 **(D)** 32

[Source: Repo: BCI_SCI Subject Test-7 (BCI/SCI Batch)]

✔ **Correct: (C) 24** — Class C has a default mask of 255.255.255.0 — the first 24 bits are the network portion, leaving 8 bits for hosts (254 usable).

Class C का default mask 255.255.255.0 होता है — पहले 24 bits network भाग होते हैं, 8 bits hosts के लिए बचते हैं (254 उपयोगी)।

★ **Nukta:** Class C = /24 network (24 bits) + 8 host bits. Class B = /16, Class A = /8.

Class C = /24 network (24 bits) + 8 host bits। Class B = /16, Class A = /8।

**Q18.** Which IP address class provides the largest number of hosts per network?

कौन-सी IP address class प्रति network सबसे अधिक hosts देती है?

**(A)** Class A **(B)** Class B
**(C)** Class C **(D)** Class D

[Source: Repo: CN Topic Test-03 (UNIT TEST-3)]

✔ **Correct: (A) Class A** — Class A reserves 7 bits for hosts (2^24 − 2 ≈ 16.7 million hosts per network) — the largest of the classes.

Class A में hosts के लिए 24 bits होते हैं (2^24 − 2 ≈ 16.7 million hosts प्रति network) — सबसे बड़ी class।

★ **Nukta:** Hosts per class: A = 2^24−2, B = 2^16−2 (65,534), C = 2^8−2 (254). D/E have no hosts (multicast/experimental).

प्रति class hosts: A = 2^24−2, B = 2^16−2 (65,534), C = 2^8−2 (254)। D/E में hosts नहीं (multicast/experimental)।

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**Q19.** Which class of IP addresses supports 65,534 hosts per network?

कौन-सी IP address class प्रति network 65,534 hosts support करती है?

**(A)** Class A **(B)** Class B
**(C)** Class C **(D)** Class D

[Source: Repo: CN Topic Test-03 (UNIT TEST-3)]

✔ **Correct: (B) Class B** — Class B has 16 host bits: 2^16 − 2 = 65,534 usable hosts per network (minus network and broadcast addresses).

Class B में 16 host bits होती हैं: 2^16 − 2 = 65,534 उपयोगी hosts प्रति network (network और broadcast address घटाने के बाद)।

★ **Nukta:** 65,534 = 2^16−2 = Class B hosts. 254 = 2^8−2 = Class C. 16,777,214 = 2^24−2 = Class A.

65,534 = 2^16−2 = Class B hosts। 254 = 2^8−2 = Class C। 16,777,214 = 2^24−2 = Class A।

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**Q20.** What is the maximum number of IP addresses in a Class C network?

Class C network में IP addresses की अधिकतम संख्या कितनी होती है?

**(A)** 254 **(B)** 256
**(C)** 1024 **(D)** 65536

[Source: Repo: CN Topic Test 5 (UNIT TEST-5)]

✔ **Correct: (B) 256** — A Class C network has 8 host bits, so 2^8 = 256 total addresses — of which 254 are usable hosts (one network ID + one broadcast address are reserved).

Class C network में 8 host bits होती हैं, इसलिए 2^8 = कुल 256 addresses — जिनमें से 254 उपयोगी hosts हैं (एक network ID + एक broadcast address reserved रहते हैं)।

★ **Nukta:** Total addresses = 2^host-bits. Usable hosts = total − 2. Exam asks BOTH — read 'addresses' vs 'hosts' carefully!

कुल addresses = 2^host-bits। उपयोगी hosts = कुल − 2। Exam दोनों पूछता है — 'addresses' और 'hosts' में ध्यान से फर्क पढ़ें!

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**Q21.** Which of the following is a Class D address?

निम्न में से कौन-सा Class D address है?

**(A)** 224.1.1.1 **(B)** 233.0.0.10
**(C)** 239.5.5.5 **(D)** All of the above

[Source: Repo: BCI Subject Test-8 (BCI/SCI Batch)]

✔ **Correct: (D) All of the above** — Class D covers 224.0.0.0-239.255.255.255 — all three addresses (224.x, 233.x, 239.x) fall in this multicast range.

Class D की range 224.0.0.0-239.255.255.255 है — तीनों addresses (224.x, 233.x, 239.x) इसी multicast range में आती हैं।

★ **Nukta:** Class D = 224-239 (multicast). Any address in this range is Class D — exam often gives mixed octets like these.

Class D = 224-239 (multicast)। इस range का कोई भी address Class D है — exam में अक्सर ऐसे मिश्रित octets दिए जाते हैं।

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**Q22.** Which class of IP address is used for multicasting?

Multicasting के लिए किस class का IP address उपयोग होता है?

**(A)** Class A **(B)** Class B
**(C)** Class C **(D)** Class D

[Source: Repo: BCI_SCI Subject Test-7 (BCI/SCI Batch) · also CN Topic Test-6]

✔ **Correct: (D) Class D** — Class D (224.0.0.0-239.255.255.255) is reserved for multicast group addresses — one packet is received by all members of the group.

Class D (224.0.0.0-239.255.255.255) multicast group addresses के लिए reserved है — एक packet group के सभी सदस्यों को मिलता है।

★ **Nukta:** D = Multicast (one-to-group). E = Experimental (240-255). Broadcast (255.255.255.255) is separate.

D = Multicast (one-to-group)। E = Experimental (240-255)। Broadcast (255.255.255.255) अलग है।

**Q23.** Which class of address is reserved for experimental purposes?
प्रायोगिक (experimental) उद्देश्यों के लिए किस class का address reserved है?

**(A)** Class C **(B)** Class B
**(C)** Class D **(D)** Class E

[Source: Repo: BCI_SCI Subject Test-7 (BCI/SCI Batch)]

✔ **Correct: (D) Class E** — Class E (240.0.0.0-255.255.255.255) is reserved for experimental/research use and is not used on the public Internet.
Class E (240.0.0.0-255.255.255.255) प्रायोगिक/अनुसंधान उपयोग के लिए reserved है और public Internet पर उपयोग नहीं होता।

★ **Nukta:** Class E = Experimental (240-255). Remember: A-ny (0), B-128, C-192, D-224 (multicast), E-240 (experiment).
Class E = Experimental (240-255)। याद रखें: A-ny (0), B-128, C-192, D-224 (multicast), E-240 (experiment)।

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**Q24.** Which IP address is reserved for future use and experimental purposes?
कौन-सा IP address भविष्य के उपयोग और प्रायोगिक उद्देश्यों के लिए reserved है?

**(A)** 127.0.0.1 **(B)** 224.0.0.1
**(C)** 240.0.0.0 **(D)** 192.168.0.1

[Source: Repo: CN Topic Test-03 (UNIT TEST-3)]

✔ **Correct: (C) 240.0.0.0** — 240.0.0.0 is the start of the Class E block (240-255) reserved for future/experimental use.
240.0.0.0, Class E block (240-255) की शुरुआत है जो भविष्य/प्रायोगिक उपयोग के लिए reserved है।

★ **Nukta:** 240.0.0.0/4 = Class E = future use. 127.x = loopback, 224.x = multicast, 192.168.x = private.
240.0.0.0/4 = Class E = future use। 127.x = loopback, 224.x = multicast, 192.168.x = private।

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**Q25.** If the number of networks and number of hosts in Class B are 2^m and (2^n − 2) respectively, then the relation between m and n is:
यदि Class B में networks और hosts की संख्या क्रमशः 2^m और (2^n − 2) है, तो m और n के बीच क्या संबंध है?

**(A)** 3m = 2n **(B)** 7m = 8n
**(C)** 8m = 7n **(D)** 2m = 3n

[Source: Repo: CN Topic Test-03 (UNIT TEST-3)]

✔ **Correct: (C) 8m = 7n** — Class B has 2^14 networks (m = 14) and 2^16 − 2 hosts (n = 16): 8 × 14 = 7 × 16 = 112, so 8m = 7n.
Class B में 2^14 networks (m = 14) और 2^16 − 2 hosts (n = 16) होते हैं: 8 × 14 = 7 × 16 = 112, अतः 8m = 7n।

★ **Nukta:** Class B classful counts: networks 2^14, hosts 2^16−2. So m=14, n=16 → 8m=7n. (Class A: 2^7 nets, 2^24−2 hosts; Class C: 2^21 nets, 2^8−2 hosts.)
Class B classful counts: networks 2^14, hosts 2^16−2। अतः m=14, n=16 → 8m=7n। (Class A: 2^7 nets, 2^24−2 hosts; Class C: 2^21 nets, 2^8−2 hosts।)

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**Q26.** Which of the following is a Class B host address?
निम्न में से कौन-सा Class B host address है?

**(A)** 230.0.0.0 **(B)** 130.4.5.6
**(C)** 230.7.6.5 **(D)** 30.4.5.6

[Source: Repo: CN Topic Test-03 (UNIT TEST-3)]

✔ **Correct: (B) 130.4.5.6** — 130.4.5.6 has first octet 130 (in 128-191) so it is Class B — and it is a valid host address in that block.
130.4.5.6 का पहला octet 130 (128-191 में) है इसलिए यह Class B है — और यह उस block का वैध host address है।

★ **Nukta:** 230.x = Class D (multicast). 30.x = Class A. 130.x = Class B — look at the FIRST octet.
230.x = Class D (multicast)। 30.x = Class A। 130.x = Class B — पहले octet को देखें।

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**Q27.** Which of the following is a valid IPv4 address?
निम्न में से कौन-सा वैध IPv4 address है?

**(A)** 192.256.1.1 **(B)** 172.31.0.1
**(C)** 300.100.50.25 **(D)** 500.10.5.2

[Source: 30 important CN MCQs (Medium blog bank)]

✔ **Correct: (B) 172.31.0.1** — Each octet must be 0-255; only 172.31.0.1 has all octets in range (the others use 256, 300 or 500).
हर octet 0-255 के बीच होना चाहिए; केवल 172.31.0.1 के सभी octets range में हैं (बाकियों में 256, 300 या 500 हैं)।

★ **Nukta:** Octet range check: 0-255 only. 172.16-172.31.x is also the private Class B range (bonus!).
Octet range check: केवल 0-255। 172.16-172.31.x private Class B range भी है (bonus!)।

**Q28.** The IP address 127.0.0.1 is used for:
IP address 127.0.0.1 किस काम में आती है?

**(A)** Broadcast **(B)** Class D network
**(C)** Loopback testing **(D)** APIPA

[Source: Repo: BCI Subject Test-8 (BCI/SCI Batch) · also CN Topic Test-03 / CN Topic Test-04]

✔ **Correct: (C) Loopback testing** — 127.0.0.1 is the loopback address — a host sends packets to it to test its own TCP/IP stack and NIC software path without leaving the machine.
127.0.0.1 loopback address है — अपने TCP/IP stack को परखने के लिए host इसे packet भेजता है, बिना machine से बाहर गए।
★ **Nukta:** Whole 127.0.0.0/8 block = loopback. `ping 127.0.0.1` tests the local IP stack.
पूरा 127.0.0.0/8 block = loopback। `ping 127.0.0.1` से local IP stack जाँचते हैं।

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**Q29.** The address 255.255.255.255 is used for:
Address 255.255.255.255 किसके लिए उपयोग होती है?

**(A)** A public IP **(B)** A default gateway
**(C)** A network ID **(D)** Broadcast to all nodes

[Source: Repo: CN Topic Test-03 (UNIT TEST-3)]

✔ **Correct: (D) Broadcast to all nodes** — 255.255.255.255 is the limited (local) broadcast address — the packet is delivered to all devices on the local network segment.
255.255.255.255 limited (local) broadcast address है — packet local network segment के सभी devices तक पहुँचता है।
★ **Nukta:** Limited broadcast = 255.255.255.255 (this network only). Directed broadcast = last address of a subnet (e.g., 192.168.1.255).
Limited broadcast = 255.255.255.255 (केवल यह network)। Directed broadcast = subnet का अंतिम address (जैसे 192.168.1.255)।

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**Q30.** What is the purpose of the reserved IP address 0.0.0.0?
Reserved IP address 0.0.0.0 का उद्देश्य क्या है?

**(A)** Default gateway **(B)** Used for DHCP servers
**(C)** Default route or 'any network' **(D)** Multicast address

[Source: Repo: CN Topic Test-03 (UNIT TEST-3)]

✔ **Correct: (C) Default route or 'any network'** — 0.0.0.0 means 'this host on this network' as a source (e.g., during boot) and, as a destination with mask /0, it is the default route (any network).
0.0.0.0 का अर्थ है 'इस network पर यह host' (source के रूप में, जैसे boot के दौरान), और /0 mask के साथ destination के रूप में यह default route (कोई भी network) है।
★ **Nukta:** 0.0.0.0/0 = default route (gateway of last resort). As source = 'not yet assigned an IP'.
0.0.0.0/0 = default route (gateway of last resort)। Source के रूप में = 'अभी IP नहीं मिला'।

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**Q31.** The IP address used by a host's packets as the source while the host is being booted (before it gets an IP) is:
जब host boot हो रहा होता है (IP मिलने से पहले) तो उसके packets का source IP address कौन-सा होता है?

**(A)** 0.0.0.0 **(B)** 1.0.0.0
**(C)** 1.1.1.1 **(D)** 255.255.255.255

[Source: Repo: CN Topic Test-04 (UNIT TEST-4)]

✔ **Correct: (A) 0.0.0.0** — Before receiving an IP (e.g., in DHCP DISCOVER), the host uses 0.0.0.0 as the source address to mean 'this host — no IP yet'.
IP मिलने से पहले (जैसे DHCP DISCOVER में) host source address के रूप में 0.0.0.0 उपयोग करता है — अर्थ 'यह host — अभी IP नहीं'।
★ **Nukta:** Boot/DHCP phase: source 0.0.0.0, destination 255.255.255.255 (broadcast). Ye DHCP handshake ka classic pattern hai.
Boot/DHCP चरण: source 0.0.0.0, destination 255.255.255.255 (broadcast)। यह DHCP handshake का classic pattern है।

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**Q32.** What is the range of addresses for APIPA (Automatic Private IP Addressing)?
APIPA (Automatic Private IP Addressing) की address range क्या है?

**(A)** 169.254.0.1 to 169.254.255.254 **(B)** 10.0.0.0 to 10.255.255.255
**(C)** 192.168.0.0 to 192.168.255.255 **(D)** 172.16.0.0 to 172.31.255.255

[Source: Repo: CN Topic Test-03 (UNIT TEST-3)]

✔ **Correct: (A) 169.254.0.1 to 169.254.255.254** — When DHCP fails, Windows hosts self-assign an APIPA address in 169.254.0.1-169.254.255.254 (link-local).
जब DHCP fail होता है, तो Windows hosts स्वयं को 169.254.0.1-169.254.255.254 (link-local) में APIPA address देते हैं।
★ **Nukta:** APIPA = 169.254.x.x (link-local, auto). Private blocks: 10.x / 172.16-31.x / 192.168.x (RFC 1918).
APIPA = 169.254.x.x (link-local, auto)। Private blocks: 10.x / 172.16-31.x / 192.168.x (RFC 1918)।

**Q33.** Which IP address can be used in a WAN (public network)?

कौन-सा IP address WAN (public network) में उपयोग हो सकता है?

**(A)** 256.0.0.1 **(B)** 172.16.0.10
**(C)** 15.1.5.6 **(D)** 127.256.0.118

[Source: Repo: CN Topic Test-03 (UNIT TEST-3)]

✔ **Correct: (C) 15.1.5.6** — 15.1.5.6 is a valid PUBLIC address (Class A). 172.16.0.10 is private (RFC 1918), 256.0.0.1 and 127.256.0.118 are invalid octets.

15.1.5.6 एक वैध PUBLIC address (Class A) है। 172.16.0.10 private (RFC 1918) है, 256.0.0.1 और 127.256.0.118 अवैध octets हैं।

★ **Nukta:** WAN needs public IPs. Private ranges (10.x, 172.16-31.x, 192.168.x) + loopback + invalid octets — sab hatana seekho.

WAN के लिए public IPs चाहिए। Private ranges (10.x, 172.16-31.x, 192.168.x) + loopback + invalid octets — सब हटाना सीखें।

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**Q34.** Which subnet mask is the default for the Class B IP address range?

Class B IP address range के लिए default subnet mask कौन-सा है?

**(A)** 255.255.255.0 **(B)** 255.0.0.0
**(C)** 255.255.0.0 **(D)** 255.255.255.255

[Source: 30 important CN MCQs (Medium blog bank) · also CN Topic Test-6]

✔ **Correct: (C) 255.255.0.0** — Class B uses a default mask of 255.255.0.0 (/16) — first 16 bits network, last 16 bits host.

Class B का default mask 255.255.0.0 (/16) होता है — पहले 16 bits network, अंतिम 16 bits host।

★ **Nukta:** Default masks: A 255.0.0.0 (/8), B 255.255.0.0 (/16), C 255.255.255.0 (/24).

Default masks: A 255.0.0.0 (/8), B 255.255.0.0 (/16), C 255.255.255.0 (/24)।

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**Q35.** Each computer connected to the internet must:

Internet से जुड़े प्रत्येक computer को क्या होना चाहिए?

**(A)** Be an IBM PC **(B)** Have a unique IP address
**(C)** Be internet compatible **(D)** Have a modem connection

[Source: SSC CGL Internet & Networking (via EduRev bank)]

✔ **Correct: (B) Have a unique IP address** — Every device on the Internet needs a unique IP address so that packets can be routed to it uniquely.

Internet पर हर device को unique IP address चाहिए ताकि packets उस तक uniquely route हो सकें।

★ **Nukta:** Unique IP = the fundamental Internet requirement (IPv4 32-bit or IPv6 128-bit).

Unique IP = Internet की मूल आवश्यकता (IPv4 32-bit या IPv6 128-bit)।

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**Q36.** Which organization is responsible for managing IP address allocation globally?

वैश्विक स्तर पर IP address allocation का प्रबंधन कौन-सा संगठन करता है?

**(A)** IEEE **(B)** ICANN
**(C)** IETF **(D)** ISO

[Source: Repo: CN Topic Test-03 (UNIT TEST-3)]

✔ **Correct: (B) ICANN** — ICANN (Internet Corporation for Assigned Names and Numbers) — through IANA/RIRs — coordinates global IP address and domain-name allocation.

ICANN (Internet Corporation for Assigned Names and Numbers) — IANA/RIRs के माध्यम से — वैश्विक IP address और domain-name allocation का समन्वय करता है।

★ **Nukta:** ICANN/IANA = addresses + names. IETF = protocols (RFCs). IEEE = LAN standards (802.x). ISO = OSI model.

ICANN/IANA = addresses + names। IETF = protocols (RFCs)। IEEE = LAN standards (802.x)। ISO = OSI model।

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**Q37.** What is the primary purpose of NAT (Network Address Translation) in IPv4?

IPv4 में NAT (Network Address Translation) का मुख्य उद्देश्य क्या है?

**(A)** To map private IP addresses to a public IP address **(B)** To increase the security of IP addresses **(C)** To facilitate communication between different subnets **(D)** All of the above

[Source: IP Addressing quiz (Studocu bank, with keys)]

✔ **Correct: (A) To map private IP addresses to a public IP address** — NAT translates private (RFC 1918) addresses to a public IP (and back) at the router, so many private hosts can share one public address.

NAT router पर private (RFC 1918) addresses को public IP में (और वापस) बदलता है, जिससे कई private hosts एक public address share कर सकें।

★ **Nukta:** NAT core = private→public mapping. Bonus effects: hides internal IPs (security), helps address exhaustion. Some banks key 'All of these' — the PRIMARY purpose is the mapping.

NAT का core = private→public mapping। अतिरिक्त प्रभाव: internal IPs छिपती हैं (security), address exhaustion कम होती है। कुछ banks 'All of these' key करते हैं — मुख्य उद्देश्य mapping ही है।

**Q38.** A host is connected to a department network which is part of a university network; the university network in turn is part of the Internet. Which is the LARGEST network among these?

एक host department network से जुड़ा है जो university network का हिस्सा है; university network बदले में Internet का हिस्सा है। इनमें सबसे बड़ा network कौन-सा है?

**(A)** The subnet to which the host belongs **(B)** The department network **(C)** The university network **(D)** The Internet

[Source: Repo: CN Topic Test-02 (UNIT TEST-2)]

✔ **Correct: (D) The Internet** — The Internet is the inter-network of all these — a network of networks — so it is the largest in the hierarchy.

Internet इन सब का inter-network — networks का network — है, इसलिए यह पदानुक्रम (hierarchy) में सबसे बड़ा है।

★ **Nukta:** Hierarchy: subnet ⊂ department LAN ⊂ university CAN ⊂ Internet. Addressing mirrors this (network prefix lengths).

पदानुक्रम: subnet ⊂ department LAN ⊂ university CAN ⊂ Internet। Addressing भी इसी तरह (network prefix lengths)।

**Q39.** Which IP address type is used to communicate with all devices on a subnet?

Subnet के सभी devices से संवाद करने के लिए किस type का IP address उपयोग होता है?

**(A)** Unicast **(B)** Broadcast **(C)** Multicast **(D)** Anycast

[Source: Repo: CN Topic Test-03 (UNIT TEST-3)]

✔ **Correct: (B) Broadcast** — Broadcast (e.g., 192.168.1.255 for a /24) delivers a packet to every device on the subnet.

Broadcast (जैसे /24 के लिए 192.168.1.255) packet को subnet के हर device तक पहुँचाता है।

★ **Nukta:** IP communication types: Unicast (one-to-one), Broadcast (one-to-all-on-subnet), Multicast (one-to-group), Anycast (one-to-nearest).

IP communication types: Unicast (one-to-one), Broadcast (one-to-all-on-subnet), Multicast (one-to-group), Anycast (one-to-nearest)।

## Section C — Subnetting & CIDR

Subnet masks, /n notation, usable hosts & subnet counts, block arithmetic — basic to advanced calculations.

**Q40.** What does the term 'subnetting' refer to in networking?

Networking में 'subnetting' शब्द का क्या अर्थ है?

**(A)** Dividing a large network into smaller logical networks **(B)** Allocating IP addresses to devices **(C)** Filtering network traffic **(D)** Assigning MAC addresses

[Source: 30 important CN MCQs (Medium blog bank) · also CN Topic Test-03]

✔ **Correct: (A) Dividing a large network into smaller logical networks** — Subnetting divides one large network into several smaller sub-networks by borrowing bits from the host part of the address.

Subnetting address के host भाग से bits उधार लेकर एक बड़े network को कई छोटे sub-networks में बाँटती है।

★ **Nukta:** Subnetting = big network → smaller networks (better performance, security, less broadcast).

Subnetting = बड़ा network → छोटे networks (बेहतर performance, security, कम broadcast)।

**Q41.** When IPv4 addressing is manually configured, which property identifies the network and host portions of the address?
जब IPv4 addressing manually configure होती है, तो कौन-सा property address के network और host भाग पहचानती है?

**(A)** DNS server address **(B)** Subnet mask
**(C)** Default gateway **(D)** DHCP server address

[Source: IP Addressing quiz (Studocu bank, with keys)]

✔ **Correct: (B) Subnet mask** — The subnet mask (e.g., 255.255.255.0) marks which bits of the IP address are the network part (1s) and which are the host part (0s).
Subnet mask (जैसे 255.255.255.0) बताता है कि IP address के कौन-से bits network भाग (1s) हैं और कौन-से host भाग (0s)।
★ **Nukta:** Mask = boundary between network and host. /24 = 255.255.255.0 = 24 network bits + 8 host bits.
Mask = network और host के बीच सीमा। /24 = 255.255.255.0 = 24 network bits + 8 host bits।

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**Q42.** In a subnet mask, the number of 1's represents:
Subnet mask में 1's की संख्या किसका प्रतिनिधित्व करती है?

**(A)** Host bits **(B)** Network bits
**(C)** Broadcast bits **(D)** Subnet bits

[Source: Repo: CN Topic Test-03 (UNIT TEST-3)]

✔ **Correct: (B) Network bits** — In a subnet mask the 1-bits mark the network (and subnet) portion; the 0-bits mark the host portion.
Subnet mask में 1-bits network (और subnet) भाग दर्शाते हैं; 0-bits host भाग दर्शाते हैं।
★ **Nukta:** Count of 1s = prefix length (/n). Example: 255.255.255.0 has 24 ones = /24.
1s की गिनती = prefix length (/n)। उदाहरण: 255.255.255.0 में 24 ones = /24।

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**Q43.** In CIDR notation, what does /16 represent?
CIDR notation में /16 का क्या अर्थ है?

**(A)** 16 host bits **(B)** 16 network bits
**(C)** 16 bytes **(D)** 16 subnets

[Source: Repo: CN Topic Test-03 (UNIT TEST-3)]

✔ **Correct: (B) 16 network bits** — The number after the slash is the count of network (prefix) bits — /16 means the first 16 bits identify the network.
Slash के बाद की संख्या network (prefix) bits की गिनती है — /16 का अर्थ पहले 16 bits network पहचानते हैं।
★ **Nukta:** CIDR /n = n network bits. Host bits = 32 − n. /16 → 2^16 − 2 hosts.
CIDR /n = n network bits। Host bits = 32 − n। /16 → 2^16 − 2 hosts।

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**Q44.** What does /24 denote in CIDR?
CIDR में /24 किसे दर्शाता है?

**(A)** 24 bits are used for the host portion **(B)** 24 bits are used for the network portion
**(C)** 24 subnets are created **(D)** 24 addresses are available in the subnet

[Source: RPSC Programmer 2024 Paper-1 (official key)]

✔ **Correct: (B) 24 bits are used for the network portion** — /24 means the first 24 bits are the network portion (mask 255.255.255.0), leaving 8 host bits.
/24 का अर्थ पहले 24 bits network भाग (mask 255.255.255.0) हैं, 8 host bits बचती हैं।
★ **Nukta:** Official RPSC 2024 key: /24 = 24 network bits. Same idea: /27 = 27 network bits, /30 = 30.
Official RPSC 2024 key: /24 = 24 network bits। वैसे ही: /27 = 27 network bits, /30 = 30।

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**Q45.** How many USABLE hosts are there in a /24 subnet?
/24 subnet में कितने उपयोगी (usable) hosts होते हैं?

**(A)** 128 **(B)** 256
**(C)** 254 **(D)** 512

[Source: Repo: CN Topic Test 5 (UNIT TEST-5) · also CN Topic Test-03]

✔ **Correct: (C) 254** — /24 leaves 8 host bits: 2^8 = 256 addresses, minus the network ID and broadcast address = 254 usable hosts.
/24 में 8 host bits बचती हैं: 2^8 = 256 addresses, network ID और broadcast address घटाने पर = 254 उपयोगी hosts।
★ **Nukta:** Formula: usable hosts = 2^host_bits − 2. /24 → 254, /25 → 126, /26 → 62, /27 → 30, /28 → 14, /29 → 6, /30 → 2.
सूत्र: उपयोगी hosts = 2^host_bits − 2। /24 → 254, /25 → 126, /26 → 62, /27 → 30, /28 → 14, /29 → 6, /30 → 2।

**Q46.** How many usable hosts are there in a /30 subnet?

/30 subnet में कितने उपयोगी hosts होते हैं?

**(A)** 2 **(B)** 4
**(C)** 6 **(D)** 14

[Source: Repo: CN Topic Test-03 (UNIT TEST-3)]

✔ **Correct: (A) 2** — /30 leaves 2 host bits: 2^2 = 4 addresses − network − broadcast = 2 usable hosts — perfect for point-to-point WAN links.

/30 में 2 host bits बचती हैं: 2^2 = 4 addresses − network − broadcast = 2 उपयोगी hosts — point-to-point WAN links के लिए एकदम सही।

★ **Nukta:** /30 = 2 hosts (router-to-router links). /29 = 6 hosts (small office).

/30 = 2 hosts (router-to-router links)। /29 = 6 hosts (छोटा office)।

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**Q47.** A subnet /30 is MOST commonly used for:

Subnet /30 सबसे अधिक किसके लिए उपयोग होती है?

**(A)** Two-host networks **(B)** Four-host networks
**(C)** Loopback only **(D)** VPN networks

[Source: Repo: BCI Subject Test-8 (BCI/SCI Batch)]

✔ **Correct: (A) Two-host networks** — A /30 gives exactly 2 usable hosts — the classic choice for point-to-point links between two routers.

/30 में ठीक 2 उपयोगी hosts मिलते हैं — दो routers के बीच point-to-point links के लिए classic विकल्प।

★ **Nukta:** Link between 2 routers = /30. WAN links ya tunnel endpoints ke liye standard.

2 routers के बीच link = /30। WAN links या tunnel endpoints के लिए standard।

---

**Q48.** A subnet mask of 255.255.255.248 allows how many usable hosts?

Subnet mask 255.255.255.248 कितने उपयोगी hosts की अनुमति देता है?

**(A)** 2 **(B)** 6
**(C)** 14 **(D)** 30

[Source: Repo: CN Topic Test 5 (UNIT TEST-5)]

✔ **Correct: (B) 6** — 255.255.255.248 = /29 → 3 host bits → 2^3 − 2 = 6 usable hosts.

255.255.255.248 = /29 → 3 host bits → 2^3 − 2 = 6 उपयोगी hosts।

★ **Nukta:** .248 = /29 = 6 hosts. Remember the /30-/29-/28 ladder: 2, 6, 14.

.248 = /29 = 6 hosts। /30-/29-/28 ladder याद रखें: 2, 6, 14।

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**Q49.** The subnet mask 255.255.255.128 provides how many subnets for a Class C network?

Subnet mask 255.255.255.128, Class C network के लिए कितने subnets देता है?

**(A)** 1 **(B)** 2
**(C)** 4 **(D)** 8

[Source: Repo: CN Topic Test-03 (UNIT TEST-3)]

✔ **Correct: (B) 2** — 255.255.255.128 borrows 1 host bit from Class C's 8: 2^1 = 2 subnets (each with 126 hosts).

255.255.255.128, Class C की 8 host bits में से 1 bit उधार लेता है: 2^1 = 2 subnets (हर एक में 126 hosts)।

★ **Nukta:** Borrowed bits n → 2^n subnets. .128 = 1 bit borrowed → 2 subnets.

उधार bits n → 2^n subnets। .128 = 1 bit उधार → 2 subnets।

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**Q50.** A network needs 5 subnets, each with at least 16 hosts. Which subnet mask (Class C) should be used?

एक network को 5 subnets चाहिए, हर एक में कम से कम 16 hosts। कौन-सा subnet mask (Class C) उपयोग करें?

**(A)** 255.255.255.192 **(B)** 255.255.255.248
**(C)** 255.255.255.240 **(D)** 255.255.255.224

[Source: Repo: CN Topic Test-03 (UNIT TEST-3)]

✔ **Correct: (D) 255.255.255.224** — 255.255.255.224 (/27) borrows 3 bits: 2^3 = 8 subnets (≥ 5) and leaves 5 host bits: 2^5 − 2 = 30 hosts (≥ 16).

255.255.255.224 (/27) 3 bits उधार लेता है: 2^3 = 8 subnets (≥ 5) और 5 host bits बचती हैं: 2^5 − 2 = 30 hosts (≥ 16)।

★ **Nukta:** Subnetting trade-off: more subnets → fewer hosts each. Borrow just enough bits (3 here).

Subnetting trade-off: ज़्यादा subnets → हर एक में कम hosts। बस पर्याप्त bits उधार लें (यहाँ 3)।

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**Q51.** A subnet mask for Class C can have how many 1's by default (remaining bits 0)?

Class C के लिए subnet mask में डिफ़ॉल्ट रूप से कितने 1's हो सकते हैं (बाकी bits 0)?

**(A)** 10 **(B)** 24
**(C)** 12 **(D)** 7

[Source: Repo: CN Topic Test-03 (UNIT TEST-3)]

✔ **Correct: (B) 24** — The default Class C mask 255.255.255.0 has 24 one-bits and 8 zero-bits.

डिफ़ॉल्ट Class C mask 255.255.255.0 में 24 one-bits और 8 zero-bits होते हैं।

★ **Nukta:** Class C default = /24 = 24 ones. (With subnetting it can go up to /30 practically.)

Class C default = /24 = 24 ones। (Subnetting के साथ यह व्यावहारिक रूप से /30 तक जा सकता है।)

**Q52.** What is the valid host range for the subnet 192.168.1.0/25?

Subnet 192.168.1.0/25 के लिए वैध host range क्या है?

**(A)** 192.168.1.0 - 192.168.1.255 **(B)** 192.168.1.1 - 192.168.1.126
**(C)** 192.168.1.1 - 192.168.1.254 **(D)** 192.168.1.0 - 192.168.1.127

[Source: Repo: CN Topic Test-03 (UNIT TEST-3)]

✔ **Correct: (B) 192.168.1.1 - 192.168.1.126** — /25 splits the block into halves of 128: the first subnet is .0-.127 — network .0, broadcast .127, so usable hosts are .1-.126.

/25 block को 128 की दो आधा-आधा भागों में बाँटता है: पहला subnet .0-.127 — network .0, broadcast .127, इसलिए उपयोगी hosts .1-.126 हैं।

★ **Nukta:** Host range = network+1 to broadcast−1. For /25: .1-.126 (first subnet) and .129-.254 (second).

Host range = network+1 से broadcast−1। /25 के लिए: .1-.126 (पहला subnet) और .129-.254 (दूसरा)।

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**Q53.** Which part of the IP address 172.16.5.4/24 identifies the network?

IP address 172.16.5.4/24 का कौन-सा भाग network पहचानता है?

**(A)** 172 **(B)** 172.16
**(C)** 172.16.5 **(D)** 172.16.5.4

[Source: Repo: CN Topic Test-03 (UNIT TEST-3)]

✔ **Correct: (C) 172.16.5** — /24 keeps the first 24 bits (three octets: 172.16.5) as the network part; .4 is the host part.

/24 में पहले 24 bits (तीन octets: 172.16.5) network भाग होते हैं; .4 host भाग है।

★ **Nukta:** Visual trick: /8 → 1 octet, /16 → 2 octets, /24 → 3 octets network.

Visual trick: /8 → 1 octet, /16 → 2 octets, /24 → 3 octets network।

---

**Q54.** A host has IP 192.168.10.45/27. Which statement is TRUE?

एक host का IP 192.168.10.45/27 है। कौन-सा कथन सही है?

**(A)** Broadcast address = 192.168.10.63 **(B)** Network ID = 192.168.10.32
**(C)** First usable host = 192.168.10.33 **(D)** All of the above

[Source: Repo: BCI Subject Test-8 (BCI/SCI Batch) · also CN Topic Test-03]

✔ **Correct: (D) All of the above** — /27 = block size 32: the subnet is .32-.63 — network .32, first host .33, broadcast .63. All three statements are true.

/27 = block size 32: subnet .32-.63 है — network .32, पहला host .33, broadcast .63। तीनों कथन सही हैं।

★ **Nukta:** Block arithmetic: 45 ÷ 32 → block 32. Network = block start; broadcast = next block − 1.

Block arithmetic: 45 ÷ 32 → block 32। Network = block की शुरुआत; broadcast = अगला block − 1।

---

**Q55.** In subnetting, reducing the number of host bits does what?

Subnetting में host bits की संख्या घटाने से क्या होता है?

**(A)** Increases the broadcast domain **(B)** Decreases the number of subnets
**(C)** Increases the number of subnets **(D)** Increases the number of hosts

[Source: Repo: CN Topic Test 5 (UNIT TEST-5)]

✔ **Correct: (C) Increases the number of subnets** — Borrowing host bits (reducing host bits) increases the number of subnets — each subnet then has fewer hosts.

Host bits उधार लेने (host bits घटाने) से subnets की संख्या बढ़ती है — हर subnet में hosts कम हो जाते हैं।

★ **Nukta:** More subnet bits ⇔ more subnets ⇔ fewer hosts each. Ye subnetting ka basic trade-off hai.

ज़्यादा subnet bits ⇔ ज़्यादा subnets ⇔ हर एक में कम hosts। यह subnetting का मूल trade-off है।

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**Q56.** In a class B subnet: IP address 125.134.112.66, mask 255.255.224.0. What is the first address (network address)?

Class B subnet में: IP address 125.134.112.66, mask 255.255.224.0। पहला address (network address) क्या होगा?

**(A)** 125.134.96.0 **(B)** 125.134.112.0
**(C)** 125.134.112.66 **(D)** 125.134.0.0

[Source: Repo: CN Topic Test-03 (UNIT TEST-3)]

✔ **Correct: (A) 125.134.96.0** — Mask 255.255.224.0 = /19 with block size 32 in the third octet: 112 = 3×32+16 → block start 96 → network 125.134.96.0.

Mask 255.255.224.0 = /19, third octet में block size 32: 112 = 3×32+16 → block की शुरुआत 96 → network 125.134.96.0।

★ **Nukta:** Block size = 256 − last non-255 octet (256−224 = 32). Floor(112/32)×32 = 96.

Block size = 256 − अंतिम गैर-255 octet (256−224 = 32)। Floor(112/32)×32 = 96।

**Q57.** A Class B network is to be split into subnets with a 3-bit subnet number. What is the maximum number of subnets and hosts in each subnet?

एक Class B network को 3-bit subnet number में बाँटना है। प्रत्येक subnet में subnets और hosts की अधिकतम संख्या कितनी होगी?

**(A)** 8 subnets and 262141 hosts **(B)** 6 subnets and 262141 hosts
**(C)** 6 subnets and 1022 hosts **(D)** None of the above

[Source: Repo: CN Topic Test-03 (UNIT TEST-3)]

✔ **Correct: (D) None of the above** — With a 3-bit subnet number: 2^3 = 8 subnets (6 usable in old classful rules) and 2^13 − 2 = 8190 hosts per subnet — none of the printed options matches.

3-bit subnet number के साथ: 2^3 = 8 subnets (पुराने classful नियमों में 6 उपयोगी) और प्रति subnet 2^13 − 2 = 8190 hosts — दिए गए विकल्पों में से कोई मेल नहीं खाता।

★ **Nukta:** Class B /16 + 3 subnet bits = /19: 8 subnets × 8190 hosts each. Since options are unusual, the correct choice is 'None of the above'.

Class B /16 + 3 subnet bits = /19: 8 subnets × 8190 hosts। विकल्प असामान्य हैं इसलिए सही विकल्प 'None of the above' है।

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**Q58.** IP address 172.16.123.213 has subnet mask 255.255.255.240. What is its network (first) address?

IP address 172.16.123.213 का subnet mask 255.255.255.240 है। इसका network (पहला) address क्या है?

**(A)** 172.16.123.240 **(B)** 172.16.123.208
**(C)** 172.16.123.213 **(D)** 172.16.123.245

[Source: Repo: CN Topic Test-03 (UNIT TEST-3)]

✔ **Correct: (B) 172.16.123.208** — 255.255.255.240 = /28 with block size 16: 213 = 13×16+5 → block start 208 → network 172.16.123.208 (range .208-.223).

255.255.255.240 = /28, block size 16: 213 = 13×16+5 → block की शुरुआत 208 → network 172.16.123.208 (range .208-.223)।

★ **Nukta:** /28 = 16-address blocks: .208-.223 here. Network = .208, broadcast = .223, hosts = .209-.222 (14 hosts).

/28 = 16-address blocks: यहाँ .208-.223। Network = .208, broadcast = .223, hosts = .209-.222 (14 hosts)।

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**Q59.** An organization has a Class B network and wishes to form subnets for 64 departments. The subnet mask would be:

एक संगठन के पास Class B network है और वह 64 departments के लिए subnets बनाना चाहता है। Subnet mask क्या होगा?

**(A)** 255.255.128.0 **(B)** 255.255.0.0
**(C)** 255.255.64.0 **(D)** 255.255.252.0

[Source: RPSC Programmer 2024 Paper-1 (official key)]

✔ **Correct: (D) 255.255.252.0** — 64 subnets need 2^6 subnets → borrow 6 host bits from Class B's 16: mask = 255.255.252.0 (/22).

64 subnets के लिए 2^6 subnets → Class B की 16 host bits में से 6 bits उधार: mask = 255.255.252.0 (/22)।

★ **Nukta:** Borrow 6 bits: 16+6 = 22 network bits = 255.255.252.0. RPSC 2024 official key.

6 bits उधार: 16+6 = 22 network bits = 255.255.252.0। RPSC 2024 official key।

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**Q60.** A Class B network has a subnet mask of 255.255.248.0. What is the maximum number of hosts per subnet?

एक Class B network का subnet mask 255.255.248.0 है। प्रति subnet hosts की अधिकतम संख्या कितनी है?

**(A)** 2046 **(B)** 1022
**(C)** 1023 **(D)** 1024

[Source: RPSC Programmer 2024 Paper-1 (official key) · Rajasthan PYQ Q56]

✔ **Correct: (A) 2046** — 255.255.248.0 = /21 → 32−21 = 11 host bits → 2^11 − 2 = 2046 usable hosts.

255.255.248.0 = /21 → 32−21 = 11 host bits → 2^11 − 2 = 2046 उपयोगी hosts।

★ **Nukta:** Official key (RPSC 2024 + Rajasthan PYQ): 2046 hosts. /21 → 2^11−2 = 2046.

Official key (RPSC 2024 + Rajasthan PYQ): 2046 hosts। /21 → 2^11−2 = 2046।

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**Q61.** An IP address 156.233.42.56 is configured with a 7-bit subnet mask (7 bits borrowed from the host part of the classful network). How many hosts and subnets are possible?

IP address 156.233.42.56 को 7-bit subnet mask (classful network के host भाग से 7 bits उधार) के साथ configure किया गया है। कितने hosts और subnets संभव हैं?

**(A)** 126 hosts, 510 subnets **(B)** 128 hosts, 512 subnets
**(C)** 510 hosts, 126 subnets **(D)** 512 hosts, 128 subnets

[Source: Unit Test -2 Solutions (OCR), p5]

✔ **Correct: (A) 126 hosts, 510 subnets** — 156.233.x is Class B (default /16). With 7 bits for HOSTS: 2^7 − 2 = 126 hosts; the remaining 9 borrowed subnet bits give 2^9 − 2 = 510 usable subnets.

156.233.x Class B है (default /16)। HOSTS के लिए 7 bits: 2^7 − 2 = 126 hosts; बची 9 subnet bits से 2^9 − 2 = 510 उपयोगी subnets।

★ **Nukta:** Bank key pattern: 7 host bits → 126 hosts; 9 subnet bits → 510 usable subnets (both −2).

Bank key pattern: 7 host bits → 126 hosts; 9 subnet bits → 510 उपयोगी subnets (दोनों −2)।

## Section D — IPv6 Addressing

128-bit addresses, address types (no broadcast), IPv6-only header changes, transition (tunneling), AAAA records.

**Q62.** In IPv6, how many bits are in the address?
IPv6 में address कितने bits का होता है?

**(A)** 32 **(B)** 64
**(C)** 128 **(D)** 256

[Source: Repo: CN Topic Test-6 (UNIT TEST-6) · Sanfoundry / Medium / MQuestions banks]

✔ **Correct: (C) 128** — IPv6 addresses are 128 bits — written as 8 groups of 4 hex digits (e.g., 2001:DB8::1).
IPv6 addresses 128 bits के होते हैं — 4 hex digits के 8 groups में लिखे जाते हैं (जैसे 2001:DB8::1)।

★ **Nukta:** Size line: IPv4 32, IPv6 128, MAC 48, Port 16. 128 bits = 2^128 ≈ 3.4 × 10^38 addresses.
Size line: IPv4 32, IPv6 128, MAC 48, Port 16। 128 bits = 2^128 ≈ 3.4 × 10^38 addresses।

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**Q63.** IPv6 does NOT use which type of address?
IPv6 किस type का address उपयोग नहीं करता?

**(A)** Broadcast **(B)** Multicast
**(C)** Anycast **(D)** None of the mentioned

[Source: Sanfoundry CN MCQs — IPv6]

✔ **Correct: (A) Broadcast** — IPv6 eliminated broadcast — it uses multicast (and anycast) for group communication instead.
IPv6 ने broadcast हटा दिया है — group communication के लिए multicast (और anycast) उपयोग होता है।

★ **Nukta:** IPv4: unicast + broadcast + multicast. IPv6: unicast + multicast + anycast (NO broadcast).
IPv4: unicast + broadcast + multicast। IPv6: unicast + multicast + anycast (broadcast नहीं)।

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**Q64.** Which features are present in IPv4 but NOT in IPv6?
कौन-सी features IPv4 में हैं लेकिन IPv6 में नहीं?

**(A)** Fragmentation by routers **(B)** Header checksum
**(C)** Options **(D)** All of the mentioned

[Source: Sanfoundry CN MCQs — IPv6]

✔ **Correct: (D) All of the mentioned** — IPv6 simplified the header: routers do not fragment packets (source does), the header checksum was removed, and the Options field became extension headers.
IPv6 ने header सरल बनाया: routers packets fragment नहीं करते (source करता है), header checksum हटा दिया गया, और Options field extension headers बन गया।

★ **Nukta:** IPv6 header = fixed 40 bytes, 8 fields, NO checksum / NO router fragmentation / NO options field.
IPv6 header = fixed 40 bytes, 8 fields, कोई checksum नहीं / कोई router fragmentation नहीं / कोई options field नहीं।

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**Q65.** Which of the following is TRUE for IPv6?
IPv6 के लिए निम्न में से क्या सही है?

**(A)** Uses 32-bit addresses **(B)** Includes a header checksum
**(C)** Supports anycast addressing **(D)** Does not support multicast

[Source: Repo: CN Topic Test 5 (UNIT TEST-5)]

✔ **Correct: (C) Supports anycast addressing** — IPv6 natively supports anycast (one-to-nearest) along with unicast and multicast.
IPv6 मूल रूप से anycast (one-to-nearest) को unicast और multicast के साथ support करता है।

★ **Nukta:** IPv6 address types: Unicast, Multicast, Anycast. No broadcast, no header checksum.
IPv6 address types: Unicast, Multicast, Anycast। Broadcast नहीं, header checksum नहीं।

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**Q66.** Which IPv6 address type is globally unique and globally routable?
कौन-सा IPv6 address type globally unique और globally routable होता है?

**(A)** Link-Local **(B)** Multicast
**(C)** Unique Local **(D)** Global Unicast

[Source: Repo: CN Topic Test 5 (UNIT TEST-5)]

✔ **Correct: (D) Global Unicast** — Global Unicast addresses (2000::/3) are the public, globally routable IPv6 addresses — like public IPv4 addresses.
Global Unicast addresses (2000::/3) public, globally routable IPv6 addresses हैं — जैसे public IPv4 addresses।

★ **Nukta:** IPv6 unicast scopes: Link-Local (fe80::/10) = this link only; Unique-Local (fc00::/7) = like private; Global (2000::/3) = Internet.
IPv6 unicast scopes: Link-Local (fe80::/10) = केवल यह link; Unique-Local (fc00::/7) = private जैसा; Global (2000::/3) = Internet।

**Q67.** IPv6 was developed by:

IPv6 किसने develop किया?

**(A)** IETF **(B)** ANSI
**(C)** ISO **(D)** IEEE

[Source: Repo: CN Topic Test-03 (UNIT TEST-3)]

✔ **Correct: (A) IETF** — The IETF (Internet Engineering Task Force) developed IPv6 (published as RFC 2460, 1998) to overcome IPv4 address exhaustion.

IETF (Internet Engineering Task Force) ने IPv6 develop किया (RFC 2460, 1998) — IPv4 address exhaustion को दूर करने के लिए।

★ **Nukta:** IPv4/IPv6 = IETF (RFCs). OSI = ISO. LAN standards = IEEE (802). ANSI = US standards body.

IPv4/IPv6 = IETF (RFCs)। OSI = ISO। LAN standards = IEEE (802)। ANSI = US standards body।

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**Q68.** Two IPv6 nodes must interoperate but are connected through intervening IPv4 routers. The best solution is:

दो IPv6 nodes को interoperable होना है लेकिन वे बीच में आने वाले IPv4 routers से जुड़े हैं। सबसे अच्छा समाधान क्या है?

**(A)** Use the dual-stack approach **(B)** Tunneling
**(C)** No solution **(D)** Replace the system

[Source: Sanfoundry CN MCQs — IPv6]

✔ **Correct: (B) Tunneling** — Tunneling wraps IPv6 packets inside IPv4 packets so they can cross the IPv4-only routers (IPv6-in-IPv4).

Tunneling IPv6 packets को IPv4 packets के अंदर लपेटती है ताकि वे IPv4-only routers पार कर सकें (IPv6-in-IPv4)।

★ **Nukta:** Transition mechanisms: Tunneling (IPv6 over IPv4), Dual-stack (both), Translation (NAT64).

Transition mechanisms: Tunneling (IPv6 over IPv4), Dual-stack (दोनों), Translation (NAT64)।

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**Q69.** Which DNS resource record should be queried to retrieve the IPv6 address associated with a name?

किसी name से जुड़ा IPv6 address पाने के लिए कौन-सी DNS resource record query करनी चाहिए?

**(A)** The AAAA record **(B)** The A record
**(C)** The NS record **(D)** The MX record

[Source: cnp3 MCQ (Bonaventure CN textbook), DNS1]

✔ **Correct: (A) The AAAA record** — The AAAA ('quad-A') record maps a domain name to an IPv6 address; the A record maps to IPv4.

AAAA ('quad-A') record domain name को IPv6 address से जोड़ती है; A record IPv4 से जोड़ती है।

★ **Nukta:** DNS records: A = name→IPv4, AAAA = name→IPv6, PTR = IP→name, MX = mail, NS = name server.

DNS records: A = name→IPv4, AAAA = name→IPv6, PTR = IP→name, MX = mail, NS = name server।

## Section E — Port Numbers, DNS/URL & Address Resolution (ARP)

16-bit ports & ranges, socket addressing, DNS records & lookups, ARP/RARP, URL layer.

**Q70.** Which of the following sets of port numbers belongs to the 'well-known' ports category?

निम्न में से कौन-से port numbers 'well-known' ports category में आते हैं?

**(A)** 80, 443, 8080 **(B)** 49152, 110, 53
**(C)** 22, 25, 80 **(D)** 1024, 3306, 8080

[Source: Repo: BCI Subject Test-8 (BCI/SCI Batch)]

✔ **Correct: (C) 22, 25, 80** — Well-known ports are 0-1023 — 22 (SSH), 25 (SMTP) and 80 (HTTP) all lie in this range; the other sets contain ports above 1023.

Well-known ports 0-1023 होते हैं — 22 (SSH), 25 (SMTP) और 80 (HTTP) तीनों इसी range में हैं; बाकी sets में 1023 से ऊपर के ports हैं।

★ **Nukta:** Port ranges: Well-known 0-1023, Registered 1024-49151, Ephemeral (dynamic) 49152-65535.

Port ranges: Well-known 0-1023, Registered 1024-49151, Ephemeral (dynamic) 49152-65535।

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**Q71.** What is the typical range of ephemeral (dynamic) ports?

Ephemeral (dynamic) ports की सामान्य range क्या है?

**(A)** 0-1023 **(B)** 1024-49151
**(C)** 49152-65535 **(D)** 100-200

[Source: Repo: BCI Subject Test-8 (BCI/SCI Batch) · also CN Topic Test 5]

✔ **Correct: (C) 49152-65535** — Ephemeral (client-side dynamic) ports are assigned from 49152-65535 by the OS for each new connection.

Ephemeral (client-side dynamic) ports OS द्वारा हर नए connection के लिए 49152-65535 से दिए जाते हैं।

★ **Nukta:** Well-known 0-1023 (services), Registered 1024-49151 (apps), Ephemeral 49152-65535 (clients).

Well-known 0-1023 (services), Registered 1024-49151 (apps), Ephemeral 49152-65535 (clients)।

**Q72.** The transport layer uses which type of addressing?

Transport layer किस प्रकार की addressing उपयोग करता है?

**(A)** IP address **(B)** MAC address
**(C)** Port number **(D)** Domain name

[Source: Repo: BCI_SCI Subject Test-7 (BCI/SCI Batch)]

✔ **Correct: (C) Port number** — The transport layer uses 16-bit port numbers for service-point addressing — to deliver data to the correct process (e.g., port 80 for a web server).

Transport layer service-point addressing के लिए 16-bit port numbers उपयोग करता है — data सही process तक पहुँचाने के लिए (जैसे web server के लिए port 80)।

★ **Nukta:** Layer addressing: Data link = MAC (48), Network = IP (32), Transport = Port (16), Application = specific/socket.

Layer addressing: Data link = MAC (48), Network = IP (32), Transport = Port (16), Application = specific/socket।

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**Q73.** Which of the following uniquely identifies a process on a host?

Host पर किसी process की unique पहचान कौन करता है?

**(A)** IP address only **(B)** Port number only
**(C)** IP address + Port number **(D)** MAC address

[Source: Repo: BCI Subject Test-9 (BCI/SCI Batch)]

✔ **Correct: (C) IP address + Port number** — A socket = IP address + port number uniquely identifies a process on a host (e.g., 192.168.1.5:80).

Socket = IP address + port number किसी host पर एक process की unique पहचान करता है (जैसे 192.168.1.5:80)।

★ **Nukta:** IP = the host (machine). IP + Port = the specific process (socket/specific address).

IP = host (machine)। IP + Port = विशिष्ट process (socket/specific address)।

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**Q74.** A URL (like www.example.com) belongs to which layer?

URL (जैसे www.example.com) किस layer से संबंधित है?

**(A)** Presentation **(B)** Application
**(C)** Transport **(D)** Network

[Source: Repo: BCI_SCI Subject Test-7 (BCI/SCI Batch)]

✔ **Correct: (B) Application** — URLs/user-friendly addresses work at the application layer — DNS resolves them to IP addresses before lower layers take over.

URLs/user-friendly addresses application layer पर काम करते हैं — DNS उन्हें IP addresses में बदलता है, फिर निचली layers काम करती हैं।

★ **Nukta:** Address levels at layers: specific (app) → port (transport) → IP (network) → MAC (data link).

Layers पर address levels: specific (app) → port (transport) → IP (network) → MAC (data link)।

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**Q75.** DNS works at which layer of the OSI model?

DNS, OSI model की किस layer पर काम करता है?

**(A)** Transport **(B)** Application
**(C)** Network **(D)** Data link

[Source: Repo: BCI_SCI Subject Test-7 (BCI/SCI Batch)]

✔ **Correct: (B) Application** — DNS is an application-layer protocol (using UDP/TCP port 53) that resolves domain names to IP addresses.

DNS एक application-layer protocol है (UDP/TCP port 53 उपयोग करता है) जो domain names को IP addresses में बदलता है।

★ **Nukta:** Application layer examples: DNS, HTTP, FTP, SMTP, SNMP, DHCP. Transport carries them (UDP 53 for DNS).

Application layer के उदाहरण: DNS, HTTP, FTP, SMTP, SNMP, DHCP। Transport उन्हें ले जाता है (DNS के लिए UDP 53)।

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**Q76.** What is the default port number for DNS?

DNS का default port number क्या है?

**(A)** 80 **(B)** 21
**(C)** 53 **(D)** 110

[Source: Repo: CN Topic Test 6 (UNIT TEST-6)]

✔ **Correct: (C) 53** — DNS uses port 53 — UDP 53 for queries and TCP 53 for large transfers (zone transfers).

DNS port 53 उपयोग करता है — queries के लिए UDP 53 और बड़े transfers (zone transfers) के लिए TCP 53।

★ **Nukta:** Port one-liners: DNS 53, HTTP 80, HTTPS 443, FTP 21/20, SMTP 25, POP3 110, Telnet 23, SSH 22, DHCP 67/68.

Port one-liners: DNS 53, HTTP 80, HTTPS 443, FTP 21/20, SMTP 25, POP3 110, Telnet 23, SSH 22, DHCP 67/68।

**Q77.** DNS is the abbreviation of:

DNS का पूरा नाम (full form) क्या है?

**(A)** Dynamic Name System **(B)** Dynamic Network System

**(C)** Domain Name System **(D)** Domain Network Service

[Source: Networking MCQ bank (MQuestions blog)]

✔ **Correct: (C) Domain Name System** — DNS = Domain Name System — the hierarchical directory that maps domain names to IP addresses.

DNS = Domain Name System — एक पदानुक्रमित directory जो domain names को IP addresses से जोड़ती है।

★ **Nukta:** DNS = Domain Name System (port 53). Hierarchical: root → TLD → authoritative servers.

DNS = Domain Name System (port 53)। पदानुक्रम: root → TLD → authoritative servers।

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**Q78.** What is the role of DNS? It is best described as:

DNS की भूमिका क्या है? सबसे सही विवरण है:

**(A)** Maps a domain name to an IP address **(B)** Translates a host's private IP to a public IP

**(C)** Translates a host's IPv6 address to an IPv4 address **(D)** Maps a MAC address to an IP address

[Source: RPSC Programmer 2024 Paper-1 (official key) · also CN Topic Test-02 / Medium bank]

✔ **Correct: (A) Maps a domain name to an IP address** — DNS resolves (maps) domain names like www.example.com to IP addresses — 'the phonebook of the Internet'.

DNS domain names (जैसे www.example.com) को IP addresses से resolve (map) करता है — 'Internet की phonebook'।

★ **Nukta:** Official RPSC 2024 key. Reverse: PTR records map IP→name. DHCP = auto IP config; NAT = private→public.

Official RPSC 2024 key। Reverse: PTR records IP→name map करती हैं। DHCP = auto IP config; NAT = private→public।

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**Q79.** A reverse lookup in DNS means a lookup to _____ using _____ records.

DNS में reverse lookup का अर्थ है _____ records का उपयोग करके _____ की lookup।

**(A)** Identify mail servers, SOA **(B)** Map an IP address to a domain name, PTR

**(C)** Map domain names to IP addresses, A **(D)** Determine the authoritative name server, NS

[Source: RPSC Programmer 2024 Paper-1 (official key) · also CN Topic Test 5]

✔ **Correct: (B) Map an IP address to a domain name, PTR** — A reverse lookup resolves an IP address back to a domain name using PTR (pointer) records — e.g., for mail-server verification.

Reverse lookup PTR (pointer) records का उपयोग करके IP address को वापस domain name में बदलती है — जैसे mail-server verification के लिए।

★ **Nukta:** Forward: name → IP (A / AAAA). Reverse: IP → name (PTR, in-addr.arpa).

Forward: name → IP (A / AAAA)। Reverse: IP → name (PTR, in-addr.arpa)।

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**Q80.** Which DNS record type maps a domain name to an IPv4 address?

कौन-सी DNS record type domain name को IPv4 address से जोड़ती है?

**(A)** CNAME **(B)** A

**(C)** MX **(D)** PTR

[Source: Repo: CN Topic Test 6 (UNIT TEST-6) · also cnp3 DNS2]

✔ **Correct: (B) A** — The A (address) record maps a domain name to its IPv4 address (AAAA does the same for IPv6).

A (address) record domain name को उसके IPv4 address से जोड़ती है (IPv6 के लिए यही काम AAAA करती है)।

★ **Nukta:** Records quick map: A = name→IPv4, AAAA = name→IPv6, PTR = IP→name, CNAME = alias, MX = mail.

Records quick map: A = name→IPv4, AAAA = name→IPv6, PTR = IP→name, CNAME = alias, MX = mail।

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**Q81.** The Address Resolution Protocol (ARP) is used for:

Address Resolution Protocol (ARP) किस काम में आता है?

**(A)** Finding the IP address that corresponds to a MAC address **(B)** Finding the MAC address that corresponds to an IP address

**(C)** Finding the IP address from the DNS name **(D)** Finding the IP address of the default gateway

[Source: RPSC Programmer 2024 Paper-1 (official key)]

✔ **Correct: (B) Finding the MAC address that corresponds to an IP address** — ARP resolves a known IP address to the unknown MAC (physical) address of the next hop on the local network.

ARP ज्ञात IP address को local network पर अगली hop के अज्ञात MAC (physical) address से जोड़ता है।

★ **Nukta:** ARP = IP→MAC (Address Resolution). RARP = MAC→IP (obsolete — replaced by BOOTP/DHCP). DNS = name→IP.

ARP = IP→MAC (Address Resolution)। RARP = MAC→IP (अप्रचलित — BOOTP/DHCP से बदला)। DNS = name→IP।

## ★ Nukte Rapid-Fire — Exam Pointers (quick revision)

One-line question (left) with its exam answer + trick (right). Revise these before the exam.

| | |
|---|---|
| MAC address size?<br>MAC address size? | ✔ 48 bits (6 bytes, 12 hex digits).<br>48 bits (6 bytes, 12 hex digits)। |
| IPv4 size? IPv6 size? Port size?<br>IPv4 size? IPv6 size? Port size? | ✔ 32 / 128 / 16 bits.<br>32 / 128 / 16 bits। |
| MAC address other names?<br>MAC address के अन्य नाम? | ✔ Physical address / hardware address / burned-in address.<br>Physical address / hardware address / burned-in address। |
| Broadcast MAC address?<br>Broadcast MAC address? | ✔ FF:FF:FF:FF:FF:FF (all 48 bits = 1).<br>FF:FF:FF:FF:FF:FF (सभी 48 bits = 1)। |
| I/G bit trick for unicast/multicast?<br>Unicast/multicast के लिए I/G bit trick? | ✔ Last bit of first octet: 0 = unicast, 1 = multicast.<br>पहले octet का अंतिम bit: 0 = unicast, 1 = multicast। |
| Class ranges by first octet (A/B/C/D/E)?<br>पहले octet से Class ranges (A/B/C/D/E)? | ✔ 0-127 / 128-191 / 192-223 / 224-239 (multicast) / 240-255 (experimental).<br>0-127 / 128-191 / 192-223 / 224-239 (multicast) / 240-255 (experimental)। |
| Default masks A/B/C?<br>Default masks A/B/C? | ✔ 255.0.0.0 /8, 255.255.0.0 /16, 255.255.255.0 /24.<br>255.0.0.0 /8, 255.255.0.0 /16, 255.255.255.0 /24। |
| Hosts per class (A/B/C)?<br>प्रति class hosts (A/B/C)? | ✔ 2^24−2 (16,777,214) / 2^16−2 (65,534) / 2^8−2 (254).<br>2^24−2 (16,777,214) / 2^16−2 (65,534) / 2^8−2 (254)। |
| Loopback address?<br>Loopback address? | ✔ 127.0.0.1 (whole 127.0.0.0/8 block).<br>127.0.0.1 (पूरा 127.0.0.0/8 block)। |
| APIPA range (DHCP fail)?<br>APIPA range (DHCP fail)? | ✔ 169.254.0.1 - 169.254.255.254.<br>169.254.0.1 - 169.254.255.254। |
| Private IP blocks (RFC 1918)?<br>Private IP blocks (RFC 1918)? | ✔ 10.x / 172.16-31.x / 192.168.x.<br>10.x / 172.16-31.x / 192.168.x। |
| 0.0.0.0 means?<br>0.0.0.0 का अर्थ? | ✔ 'This host — no IP yet' (source) or default route (0.0.0.0/0).<br>'यह host — अभी IP नहीं' (source) या default route (0.0.0.0/0)। |
| Usable hosts formula?<br>Usable hosts सूत्र? | ✔ 2^host-bits − 2. /24→254, /25→126, /27→30, /30→2.<br>2^host-bits − 2। /24→254, /25→126, /27→30, /30→2। |
| Subnets formula?<br>Subnets सूत्र? | ✔ 2^borrowed-bits. 3 bits → 8 subnets.<br>2^borrowed-bits। 3 bits → 8 subnets। |
| CIDR /24 means?<br>CIDR /24 का अर्थ? | ✔ 24 network bits (mask 255.255.255.0).<br>24 network bits (mask 255.255.255.0)। |
| /30 used for?<br>/30 किसके लिए? | ✔ Point-to-point links (2 usable hosts).<br>Point-to-point links (2 उपयोगी hosts)। |

| | |
|---|---|
| Subnetting trade-off?<br>Subnetting trade-off? | ✔ More subnets ⇔ fewer hosts each.<br>ज़्यादा subnets ⇔ हर एक में कम hosts। |
| IPv6 has broadcast?<br>IPv6 में broadcast है? | ✔ NO — IPv6 uses multicast + anycast, no broadcast.<br>नहीं — IPv6 multicast + anycast उपयोग करता है, broadcast नहीं। |
| IPv6 transition techniques?<br>IPv6 transition techniques? | ✔ Tunneling (IPv6-in-IPv4), Dual-stack, Translation (NAT64).<br>Tunneling (IPv6-in-IPv4), Dual-stack, Translation (NAT64)। |
| Well-known / registered / ephemeral port ranges?<br>Well-known / registered / ephemeral port ranges? | ✔ 0-1023 / 1024-49151 / 49152-65535.<br>0-1023 / 1024-49151 / 49152-65535। |
| Port numbers: DNS, HTTP, HTTPS, FTP, SMTP, SSH?<br>Port numbers: DNS, HTTP, HTTPS, FTP, SMTP, SSH? | ✔ 53, 80, 443, 21, 25, 22.<br>53, 80, 443, 21, 25, 22। |
| Socket = ?<br>Socket = ? | ✔ IP address + Port number (identifies a process).<br>IP address + Port number (process की पहचान)। |
| DNS records: A / AAAA / PTR / MX / CNAME?<br>DNS records: A / AAAA / PTR / MX / CNAME? | ✔ name→IPv4 / name→IPv6 / IP→name / mail / alias.<br>name→IPv4 / name→IPv6 / IP→name / mail / alias। |
| ARP vs RARP?<br>ARP vs RARP? | ✔ ARP = IP→MAC. RARP = MAC→IP (obsolete, use DHCP).<br>ARP = IP→MAC। RARP = MAC→IP (अप्रचलित, DHCP उपयोग करें)। |
| NAT purpose?<br>NAT उद्देश्य? | ✔ Map private IPs → public IP (save addresses + hide internals).<br>Private IPs → public IP map करना (addresses बचाना + internals छिपाना)। |
| Per-hop vs end-to-end in a routed packet?<br>Routed packet में per-hop vs end-to-end? | ✔ MAC changes per hop; IP addresses stay end-to-end.<br>MAC हर hop पर बदलता है; IP addresses end-to-end रहते हैं। |
| Who allocates IP addresses globally?<br>Global IP addresses कौन allocate करता है? | ✔ ICANN/IANA (with RIRs). IETF makes protocols, IEEE makes LAN standards.<br>ICANN/IANA (RIRs के साथ)। IETF protocols बनाता है, IEEE LAN standards। |